Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/10/2005
PI Name:	Pickering, Karen D Ph.D.		
Project Title:	Advanced Water Recovery Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2688
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	EC3		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	01/01/1998	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Edeen, Marybeth		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This task provides for technology development for water recovery systems. The task encompasses both biological and physical/chemical system development. Together these technology areas provide technology development for both near and long-term ISS requirements as well as technology development for future extended duration missions. The primary focus is on microgravity-compatibile technologies and ISS-similar wastewater streams; however, early definition of more complex wastewater streams is planned. Included in the biological system development is continued evaluation of bench-scale biological water processors and testing of microgravity-compatible nitrification systems. Physical-chemical system development includes development of high-recovery membrane filtration systems and optimization of postprocessing systems.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Water supplies are a significant consumable in a spacecraft life support system. Efficient technologies for water recovery and reuse greatly reduce the cost of spacecraft resupply. This task develops technology for water recovery systems, including biological systems, membrane filtration systems, and photolytic post-processing systems. Establishing the performance and reliability of these systems is the initial step in reducing the cost of resupply for spacecraft water systems.
Task Progress:	An AWRS technology development project continued this year to study the minimum size anoxic reactor to treat a given flight crew's wastewater for a specific time period. AWRS Development Facility Anoxic Biological Water Processor test stands operated throughout FY05, with nitrification and denitrification bioreactors successfully coupled in June. Good removal of total organic carbon (TOC) has been achieved through the test points to date, although performance of the nitrifier has been below expectations. Also within AWRS, work is on-going to collect sufficient data to accurately size a minimum mass water Post Processing System (PPS). PPS Phase II testing, which was to provide data which can be used to accurately size a photolytic subassembly, was successfully completed in January. The PPS test stand was then reconfigured for Phase III testing, which was to compare the performance of an optimized PPS versus a system approximating the JSC Integrated Water Recovery System Test post processor. PPS Phase III testing was successfully concluded in August. During the course of the Phase III testing, the PPS test stand was successfully operated for 68 days of near-continuous batch operation and processed over 700 liters of influent. Phase III data is currently being compiled and will be assessed in an interim report. In the area of physical/chemical systems, the Cascade Distillation Subsystem (CDS) project was initiated. This project will develop and demonstrate an advanced wastewater processing subsystem that is based on centrifugal vacuum distillation with a team comprised of commercial, government, and academic partners. The CDS utilizes an innovative and proven multi-stage thermodynamic process to efficiently produce purified water, and its rotary centrifugal design provides gas/liquid phase separation and liquid transport (pumping) under microgravity conditions. In FY05, CDS breadboard testing was completed and a milestone was achieved with the completion of the CDS prototype design review in November. In the area of AWRS Development Facility enhancements, a second generation Waste Water Collection and Transportation System (WWCTS) was completed in September. The WWCTS is for the collection and transport of human waste water for the evaluation of water processing systems. WWCTS acceptance testing was finished at the end of September.
Bibliography Type:	Description: (Last Updated: 09/21/2006)
Papers from Meeting Proceedings	Muirhead D, Potess MD. "Experiences with a Space and Science Club in a Low-Income Community in South Texas." 35th International Conference on Environmental Systems and 8th European Symposium on Space Environmental Control Systems, July 2005. SAE Technical Paper 2005-01-3101, July 2005. , Jul-2005
Papers from Meeting Proceedings	Vega L, Muirhead D. "Fate of Dissolved and Particulate Carbon in an Anoxic Biological Water Processor." 35th International Conference on Environmental Systems and 8th European Symposium on Space Environmental Control Systems, July 2005. SAE Technical Paper 2005-01-3019, July 2005. , Jul-2005
Papers from Meeting Proceedings	Haddock D, Muirhead D. "Inline Monitoring and Evaluation of Inorganic Gases from a Nitrification Membrane Bioreactor." 35th International Conference on Environmental Systems and 8th European Symposium on Space Environmental Control Systems, July 2005. SAE Technical Paper 2005-01-3021, July 2005. , Jul-2005